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A novel method to measure the semantic similarity of HPO terms

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***Abstract*—It is critical yet remains to be challenging to make precise disease diagnosis from complex clinical features and highly heterogeneous genetic background. Recently, phenotype similarity has been effectively applied to model patient phenotype data. However, the existing measurements are revised based on the Gene Ontology-based term similarity models, which are not optimized for human phenotype ontologies. We propose a new similarity measure called $PhenoSim$. Our model includes a noise reduction component to model the noisy patient phenotype data, and a path-constrained Information Content-based method for phenotype semantics similarity measurement. Evaluation tests compared $PhenoSim$ with four existing approaches. It showed that $PhenoSim$ could effectively improve the performance of HPO-based phenotype similarity measurement, thus increasing the accuracy of phenotype-based causative gene prediction and disease prediction.**

***Index Terms*—Human Phenotpe Ontology; Semantic Similarity; Phenotype Similarity; Noise Reduction; Causative Gene Prediction; Disease Prediction.**

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I. Introduction

In the last five years, Mendelian disease and cancer diagnosis have been significantly accelerated by the rapidly developing next generation sequencing (NGS) techniques (such as whole genome sequencing and whole exome sequencing) [1], [2], [3]. However, purely sequence-based clinical disease diagnosis remains challenging for many other diseases with complex phenotypes and high genetic heterogeneity. This is mainly because of the difficulty of understanding and modeling the genetic variants related to complex patient phenotypic features [4].

Patient phenotypes are usually defined as the observable characteristics of patients above the molecular level, such as anatomy, behavior, and biomedical properties [5]. Tools that bridge the genetic variances and biological process activities with advanced phenotype data analysis have played a central role in deciphering gene or pathway functions in life science research [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16]. A key step in those tools is to precisely measure phenotypic features, and incorporate such information into the framework of clinical diagnosis to improve clinical diagnosis efficiency. To this end, a structured and controlled vocabulary, such as ontology, is often required.

Ontologies have been demonstrated in many cases to be informative for representing knowledge as terms and their relationships with a directed acyclic graph (DAG) [17], [18], [19], [20], [21], [22]. Since 2008, Robinson *et al* have constructed and maintained an ontology namely Human Phenotype Ontology (HPO) to describe human phenotypic abnormalities that have been encountered in human disease [5]. Nowadays, HPO has become the most popular resource for providing a structured and controlled vocabulary to unify the representation of phenotypic features involved in human diseases [23], [24], [25], [26]. HPO is often integrated with NGS data to aid disease diagnosis [27], [28], [29].

To improve diagnostic efficiency, computational tools have

been developed to quantify the phenotypic similarity between patient symptoms and curated historical disease data or known phenotypes related with a gene [30], [31], [32]. Among them, computing HPO-based phenotype similarity plays a critical role in completing disease diagnosis process.

In literature, tools such as Phenomizer[30], OWLSim [33] and HPOSim [32] have been developed to exploit HPO-based semantic similarity. Several of them borrow ideas from Gene Ontology (GO) based semantic similarity approaches, which have been extensively studied and widely used in the last decade [34], [35], [36], [37], [38], [39], [40], [41].

Phenomizer and Masino *et al.* utilizes information content (IC) to calculate the HPO-based semantic similarity between any two phenotype ontology terms [30], [31]. The IC of a term represents the specificity of the term. The terms at a lower level of HPO tend to have higher IC, and vise verse. The IC of two phenotype terms is the lowest common ancestor of the two terms in the HPO structure. Mathematically, given two HPO terms t_1 and t_2, let t_{LCA} represent their lowest common ancestor, the similarity of t_1 and t_2 is calculated as follows:

$$Sim_{IC}(t_1, t_2) = IC(t_{LCA}) = -log\frac{|D_{t_{LCA}}|}{|D|} \quad (1)$$

where $D_{t_{LCA}}$ and D represent the set of diseases annotated by t_{LCA} and the set of all the annotated diseases in HPO annotation database, respectively. An evaluation test shows that this approach outperforms the term-matching approaches that do not consider the semantic relationships between terms [30].

Based on the IC-based measurement, several other methods have been proposed to calculated group-wise phenotype semantic similarity. For example, PhenomeNet [42] and OWLSim [33] employ simGIC [43] to calculate the similarity between two sets of phenotype terms. Mathematically, given two sets of terms T_1 and T_2, their similarity is calculated as follows.

$$Sim_{GIC}(T_1, T_2) = \frac{\sum_{t \in T_1 \cap T_2} IC(t)}{\sum_{t \in T_1 \cup T_2} IC(t)} \quad (2)$$

HPOSim [32] implements several semantic similarity approaches to calculate the phenotype similarities, such as Jiang [44] and Schlicker[45]. HPOSim can provide useful functions for disease/gene comparison based on HPO.

While the aforementioned approaches have been widely used in clinical research, they calculate phenotype semantic similarities based on the designs optimized for measuring GO-based semantic similarity without taking the unique properties of HPO into account. First, the biological meaning of the HPO structure is different with that of GO. While low-level sibling terms in GO are often considered to be similar to each other, we cannot simply assume that sibling terms in HPO have any associations at the gene level or share any disease symptoms, no matter whether the terms are at the low level or close to the root term of HPO. For example, terms "Split hand (HP:0001171)" and "Areflexia of upper limbs (HP:0012046)" are two leaf terms in HPO, but between them, there is no known gene-level associations nor shared disease symptoms. Second, patient phenotypes are in general not well recognized and annotated. Phenotype term measurement could be greatly hindered by the high noises in the patient phenotype data [31]. It is necessary to model the phenotype noise when calculating phenotype similarities.

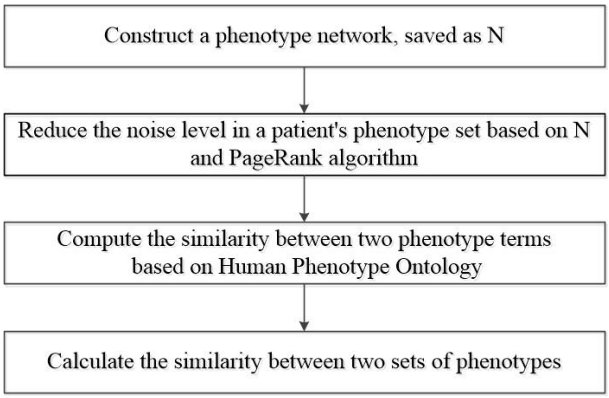


Fig. 1: The workflow of PhenoSim.

In this article, we present a new approach called $PhenoSim$ to calculate the phenotype semantic similarity based on HPO. Comparing with the existing approaches, $PhenoSim$ has the following advantages:

- To the best of our knowledge, $PhenoSim$ is the first semantic similarity approach that is specially optimized for HPO;
- We develop a novel path-constrained Information Content (IC) to calculate the similarity between two HPO terms;
- $PhenoSim$ constructs a phenotype network and exploits a PageRank-based method to model the noises in the patient phenotype data set.

II. Methods

We propose $PhenoSim$, a new phenotype semantic similarity measurement optimized for phenotype ontologies (specifically HPO). $PhenoSim$ has four steps. First, it constructs a phenotype network N using phenotype ontologies and gene-phenotype associations. Second, given T, a set of clinical phenotypes of a patient, it filters noises based on N using PageRank [46] and saves the results in T^k. Third, it computes the similarity between two phenotype terms t_1 and t_2. Finally, it computes the similarity between T_1^k and T_2^k, which are the corresponding phenotype sets of patients p_1 and p_2. The diagram of the whole process is shown in Figure 1.

A. Phenotype network construction

HPO provides a structured and controlled vocabulary to describe phenotypes and the genes associated with the phenotypes [5]. The HPO term to gene associations is mainly maintained at the OMIM database [47]. It is generally understood that the phenotype terms associated with the same genes are closely related to each other at the molecular level [48]. Hence, we identify the relationships between HPO terms using

the genes associated with them. Mathematically, given two phenotype terms t_1 and t_2, let G_1 and G_2 be the sets of their associated genes respectively. We adopt the Jaccard Index [49] to calculate the association between t_1 and t_2 based on their associated genes:

$$Sim(t_1, t_2) = \frac{|G_1 \cap G_2|}{|G_1 \cup G_2|} \tag{3}$$

Function $Sim(t_1, t_2)$ ranges between 0 and 1, and a large value indicates that terms t_1 and t_2 are similar.

Based on the pair-wise association scores for all the phenotype terms in HPO calculated using Equation 3, we construct a phenotype network $N(V, E)$ where nodes in V represent phenotype terms, and two nodes are directly connected if the association score between them is larger than a user-given threshold (in our experiments, 0), and the edge weight is the association score computed using Equation 3.

B. Phenotype network noise reduction

It is technically challenging to precisely recognize all the patient phenotypes at the data collection step. Therefore, the noises in the patient phenotype data of HPO cannot be simply ignored [31]. To this end, we develop a new approach to reducing the noise level in patient's phenotype set P.

Given a patient's phenotype term set T, subnetwork of $N(V, E)$ called $N_T(T, E')$ can be generated using the approach in the previous subsection ($E' \subset E$, $T \subset V$). For a given disease, its corresponding correctly recognized phenotype terms are high similar to each other, in that their associated gene groups are highly overlapped [48]. Thus, we assume that in N_T the correctly recognized phenotype terms are the *important* nodes, and the associations between them are *high*. Based on the assumption, we differentiate the correctly recognized phenotype terms of a patient from noises using network topological properties such as node centrality, an index to describe the node significance in a network [50].

To compute node centrality, we adopt the PageRank algorithm [46]. Let $M_{n\times n}$ be the adjacent matrix of the subnetwork N_T, where n is the number of phenotypes in N_T, and each element m_{ij} in $M_{n\times n}$ is the phenotype similarity value between phenotype t_i and t_j computed in the previous subsection. Its value is 0 if phenotype t_i and t_j are not directly connected in N_T.

Each phenotype similarity value m_{ij} in $M_{n\times n}$ is divided by the sum of all the similarity values in column j, which makes the sum of each column always be 1. The adjusted adjacent matrix is saved as M'. With M', we iteratively update the probability vector p using Equation 4:

$$p_i = \alpha M' p_{i-1} + (1 - \alpha)p \tag{4}$$

where p is the initial vector defined as $p = \frac{1}{n}[1, 1, \ldots, 1]_n^T$. α is the damping factor, which is a user given threshold. p_i is the probability vector after i iterations. Particularly, $p_0 = p$.

A stationary vector can be obtained after certain iterations. The iteration stops if the distance of two vectors is less than a small parameter. The distance between vector p_i and p_{i-1} is calculated using

$$dist(p_i, p_{i-1}) = \sum_{j=0}^{n-1} |p_i[j] - p_{i-1}[j]| \tag{5}$$

where $p_i[j]$ is the jth element in vector p_i.

Finally, all the phenotypes in a patient's phenotype term set T are ranked based on the corresponding probability in the stationary vector. The top k phenotypes with the highest probabilities are selected as the well recognized phenotypes of the patient, denoted as T^k.

C. Measuring phenotype similarity

Sibling terms in the HPO structure are not necessary to have strong associations at gene level (please refer the example in the Introduction section). Alternatively, semantically similar HPO terms are often "reachable", i.e., if two HPO terms t_1 and t_2 are similar, then there highly likely exists a directed path from one term to the other in the directed acyclic graph of HPO. Therefore, we define a new HPO term semantic similarity measurement as:

$$sim(t_1, t_2) = \begin{cases} \min(IC(t_1), IC(t_2)) & reachable \\ 0 & otherwise \end{cases} \tag{6}$$

where $IC(t)$ is the information content of phenotype term t, defined as $IC(t) = \ln \frac{U}{U_t}$, where U and U_t are the number of annotations associated with the root term and t, respectively (the annotations associated with all their descendants are also included). If t_1 and t_2 are reachable, the similarity is the minimum of their information contents. If t_1 and t_2 are unreachable, the similarity is 0.

D. Calculating phenotype set similarity

It is often required to predict whether a patient has certain disease or disease related gene. To this end, it is necessary to compare the phenotype set of a patient to the all the phenotypes associated to a disease or a disease related gene. While the patient phenotype set can be obtained in clinical treatment, the latter one are available in public databases such as OMIM [47].

Given a patient p_1 and a gene (or disease), let T_1^k and T_2^k be their associated phenotype term sets. T_1^k is the result of the noise reduction process in the previous subsection. T_2^k is set of phenotypes corresponding to the gene (or disease) obtained from the HPO database. We calculate the semantic similarity between the two patients based on the aggregation of the pair-wise similarities between terms across T_1^k and T_2^k by adopting the measure in [31].

$$Sim_{set}(T_1^k \to T_2^k) = \frac{1}{N_1} \sum_{t_i \in T_1^k} \max_{t_j \in T_2^k} sim(t_i, t_j) \tag{7}$$

$$Sim_{set}(T_2^k \to T_1^k) = \frac{1}{N_2} \sum_{t_j \in T_2^k} \max_{t_i \in T_1^k} sim(t_i, t_j) \tag{8}$$

where $sim(t_i, t_j)$ is the phenotype similarity calculated using Equation 6. N_1 and N_2 are the size of phenotype set T_1 and T_2 respectively. Note that since Equation 7 and 8 are asymmetric, the output depends on the order of the input. To avoid the asymmetry result, the similarity of two phenotype sets are calculated as:

$$Sim_{sym}(T_1^k, T_2^k) = \frac{1}{2}\left(Sim_{set}(T_1^k \rightarrow T_2^k) + Sim_{set}(T_2^k \rightarrow T_1^k)\right) \tag{9}$$

The pseudocode for calculating the similarity between two sets of phenotypes is shown in Algorithm 1.

Algorithm 1 Symmetric Term-set Semantic Similarity

```
procedure SYMMETRIC TERM-SET SEMANTIC SIMILARITY(T1,T2)
Require: T1: phenotype set 1; T2: phenotype set 2
Ensure: Sim_sym: similarity between T1 and T2
    Sim_sym ← 0; Sim_1 ← 0; Sim_2 ← 0; max ←0;
    for all t_i ∈ T1 do
        max ← 0
        for all t_j ∈ T2 do
            tempSim = sim(t_i, t_j)
            if tempSim ≥ max then
                max ← tempSim
            end if
        end for
        Sim_1 = Sim_1 + max
    end for
    Sim_1 = Sim_1/N_1
    for all t_i ∈ T2 do
        max ← 0
        for all t_j ∈ T1 do
            tempSim = sim(t_i, t_j)
            if tempSim ≥ max then
                max ← tempSim
            end if
        end for
        Sim_2 = Sim_2 + max
    end for
    Sim_2 = Sim_2/N_2
    Sim_sym = (Sim_1 + Sim_2) × 0.5
end procedure
```

III. Results

A. Data preparation

The Human Phenotype Ontology (HPO) data were downloaded from the HPO official website (http://human-phenotype-ontology.github.io/) on July 4th, 2014. It includes $61,784$ phenotype-gene relationships and $99,186$ phenotype-disease relationships. $PhenoSim$ was implemented with Java SDK 7 and the JUNG library [51].

For performance evaluation, we first generated simulated patients based on the curated disease phenotype feature set used in [31]. In this dataset, for each of the 33 selected diseases, its disease causative genes, associated phenotypes, and penetrance of each phenotype are available. The patient simulation process is as follows. First, we randomly assign a disease to each patient. Second, for a given patient, we generated a random number between 0 and 1 (followed standard uniform distribution) for every phenotype associated with the assigned disease. If the random number was smaller than the penetrance value of the phenotype, the phenotype was assigned to the patient. Each simulated patient must have at least one phenotype. This set is named as the optimal phenotype set. We repeated the process for 100 times. As a result, $3,300$ simulated patients called "optimal patients with known causative genes", were generated.

The second evaluate data set is a simulation of the real clinical data. In the real clinical practice, patient phenotype sets often contain noise. Therefore, based on the optimal set, we generated a simulated patient set with added noise. Specifically, for every disease d, we randomly generated a large set of noise phenotype terms with the criterion that they (and their descendants) do not associate with any of the causative genes associated with d. For a given patient with disease d, we randomly selected noise phenotype terms of d and added them to the patient phenotype term set T, such that the number of noise terms is half of the optimal terms in the first dataset. Particularly, if a patient only had one optimal phenotype term, no noise term was added. Finally, $3,300$ simulated patients with noisy phenotype terms, called "noisy patient data with known causative genes", were simulated. In the dataset, for each simulated patient, there are in average 7.74 phenotype terms. The phenotype terms distribution is shown in Additional file 1.

Third, we simulated patient sets with known diseases using data from OMIM [47]. The simulation process is the same as the aforementioned method except for the criterion for selecting noise phenotype terms. We required that the noisy terms and their descendants do not associate with any of the known diseases of the simulated patient. We simulated 100 patients for each of the 240 diseases (more than 30 HPO-term annotations in OMIM). Finally, datasets "noisy patient data with known diseases" and "optimal patient data with known disease", each having $24,000$ simulated patients, were generated. In the former dataset, the number of phenotype terms associated with a simulated patient ranges between 1 and 120, and the averaged number is 18.37. The phenotype term distribution is shown in Additional file 2.

B. Performance evaluation on causative gene prediction

We adopted the evaluation criterion from [31] to test whether the causative genes of a patient can be computationally identified. In this experiment, T_1^k and T_2^k are the phenotype sets corresponding to a simulated patient and a gene respectively. For a given patient, we computed the similarity score between every gene and the patient using $PhenoSim$, and then rank all the genes by their similarity scores from the largest to the smallest. If the causative gene's rank is higher than any other genesis, we conclude that $PhenoSim$

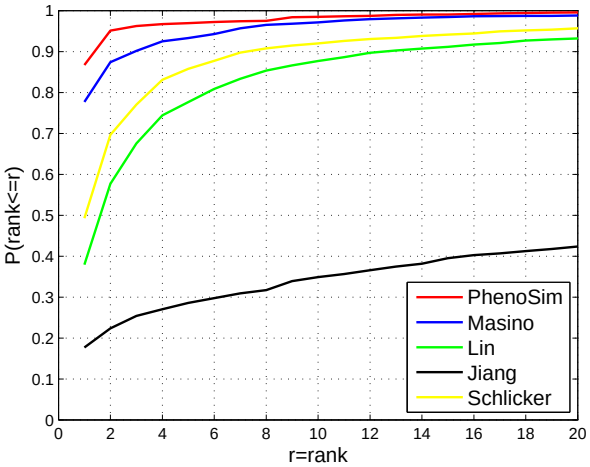


Fig. 2: Cumulative distribution of the rank of the causative genes on the "noisy patient data with known causative genes" dataset. The x-axis is the threshold for the causative gene rank. The y-axis is the ratio of patients satisfying the ranking threshold.

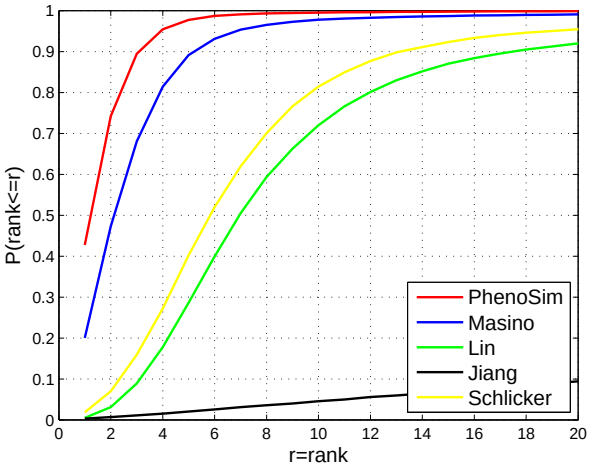


Fig. 3: Cumulative distribution of the rank of the patient-associated diseases on the "noisy patient data with known diseases" dataset. The x-axis is the threshold for the disease rank. The y-axis is the ratio of patients satisfying the ranking threshold.

can accurately predict the causative gene. Similarly, we test the performance of four existing approaches, i.e. Masino [31], Lin [52], Jiang [44], and Schlicker [45], on the datasets described above.

On the "noisy patient data with known causative genes" dataset, we tested all the five method on a set of $2,488$ available genes that have at least one HPO term annotation. The result shows that $PhenoSim$ performed the best in all the five methods (Figure 2). On 86.72% simulated patients, their causative genes are ranked the highest when $PhenoSim$ is applied. In comparison, the percentages of the highest ranked causative genes using Masino, Lin, Jiang, and Schlicker methods are 77.69%, 37.92%, 17.67% and 49.26% respectively. On 98.48% of simulated patients, the causative genes are ranked among top 10 using $PhenoSim$, while the percentages using Masino, Lin, Jiang, and Schlicker methods are 97.12%, 87.69%, 34.89% and 91.97% respectively. Furthermore, Figure 2 shows that the causative gene constantly ranks significant higher on $PhenoSim$ than on the other methods if a high-rank threshold (r) is applied. It indicates that $PhenoSim$ could be potentially helpful to narrow down the causative gene candidate set in practical clinical studies.

In addition, we also evaluated $PhenoSim$ on the "optimal patient data with known causative genes" dataset. The result shows that $PhenoSim$ and Lin perform best in all compared methods (Additional file 3). Note that we can not use the OMIM-based datasets in this test, because the causative genes of the diseases in OMIM are largely unknown [47].

C. Performance evaluation on disease prediction

We adopted the evaluation criterion from [31] to test whether the disease of a patient can be computationally identified. In this experiment, T_1^k and T_2^k are phenotype sets corresponding to a simulated patient and a disease respectively. For a given patient, we computed the similarity score between every disease and the patient using $PhenoSim$, and then rank all the diseases by their similarity scores from the largest to the smallest. If the patient-associated disease's rank is higher than any other genesis, we conclude that $PhenoSim$ can accurately predict the disease of the patient.

On the "noisy patient data with known disease" dataset, we tested all the five methods on $2,552$ diseases appeared in both HPO and OMIM. The result shows that $PhenoSim$ performed the best in all the five methods (Figure 3). The patient-associated diseases are ranked the highest on 42.74% of the patients if $PhenoSim$ is applied. In comparison, the percentages using Masino, Lin, Jiang, and Schlicker methods are 20.04%, 0.50%, 0.32% and 1.82% respectively. If we relax the criterion from the top rank to top-5, the precision of $PhenoSim$ is 97.72% (8.58% higher than the second best method Masino), while the percentages using Masino, Lin, Jiang, and Schlicker methods are 89.14%, 28.61%, 2.02% and 40.20% respectively. Furthermore, we found that the performance increases steadily with the increase of the number of phenotype terms associated with a patient (Additional file 4), indicating that rich phenotype annotations can improve the precision of phenotype-based disease diagnosis.

We evaluated $PhenoSim$ on the "optimal patient data with known disease" dataset, and the result shows that $PhenoSim$ performs the best among all the compared methods (Additional file 5). We also apply $PhenoSim$ on the two "patients with known causative genes" datasets for disease prediction [31], and the results show that $PhenoSim$ outperforms all the other methods on both the optimal and noisy sets (Additional file 6 and 7).

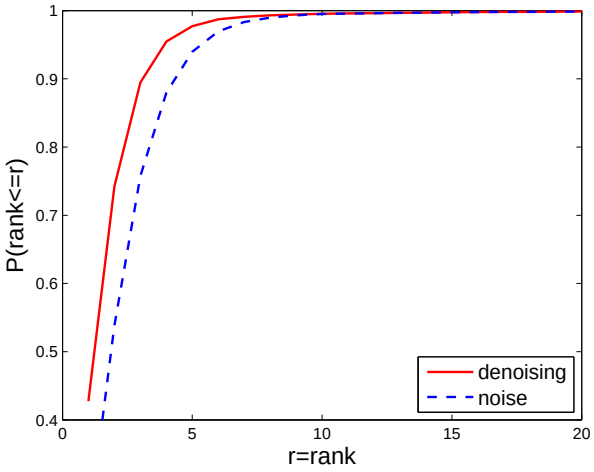
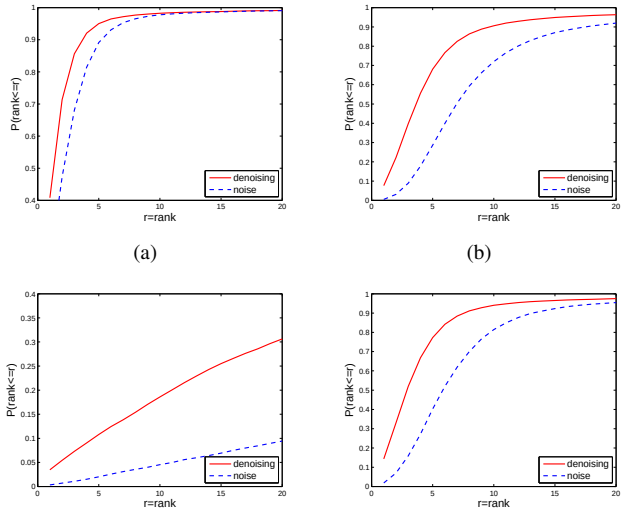


Fig. 4: Comparison of the cumulative disease-ranking distributions of the original $PhenoSim$ and the revised $PhenoSim$ without noise reduction. The x-axis is the ranking threshold. The y-axis is the ratio of patients satisfying the ranking threshold. The red line and blue dash line represent the methods with and without noise reduction respectively.



(a) (b) (c) (d)

Fig. 5: Comparison of cumulative distribution of the disease rank for Masino (a), Lin (b), Jiang (c) and Schlicker (d) methods with and without noise reduction. The x-axis is the ranking threshold. The y-axis is the ratio of patients satisfying the ranking threshold. The red line and blue dash line represent the methods with and without noise reduction, respectively.

D. Effectiveness of noise reduction

Network-based noise reduction is one of the key components of $PhenoSim$. To test whether this step can significantly affect the overall performance, we revised $PhenoSim$ by removing the noise reduction component, and compared the results of it with that of the original $PhenoSim$ on the "noisy patient data with known disease" dataset used in the "Performance evaluation of disease prediction" section. We chose this dataset because of the rich optimal and noisy phenotypes in this dataset.

The result shows that the noise reduction component can increase the performance of $PhenoSim$ (Figure 4). Using the original $PhenoSim$, the patient-associated disease ranks among top-5 on 97.72% of the patients. Removing the noise reduction reduces the percentage to 93.96%.

Furthermore, we test the availability and expandability of the noise reduction component, we applied it on the four compared methods. Results show that the noise reduction component can significantly improve the performance of all the four methods (Figure 5). On Masino, the percentage of simulated patients, whose associated diseases rank among top-5, increased from 89.14% to 95.02% (Figure 5(a)). Similarly, for Lin, Jiang and Schlicker methods, the percentages increased from 28.61% to 68.05% (Figure 5(b)), from 2.02% to 10.79% (Figure 5(c)) and from 40.20% to 77.28% (Figure 5(d)) respectively. In conclusion, noise reduction component of $PhenoSim$ can be generally applied to improve the accuracy of a phenotype similarity measurement.

IV. Conclusion

Recently, next generation sequencing techniques have significantly accelerated disease diagnosis. However, for many diseases with complex phenotypes and high genetic heterogeneity, the disease diagnosis remains challenging. Hence, HPO-based phenotype similarity could be a powerful tool to effectively accelerate the disease diagnosis process. In this article, we proposed a novel method called $PhenoSim$ to measure the phenotype semantic similarity by using a path-constrained Information Content based method. By well-modeling the noises in patient phenotype datasets, $PhenoSim$ outperforms four existing approaches on all the four patient datasets on causative gene prediction and disease prediction, indicating that $PhenoSim$ could be potentially helpful to narrow down the causative gene or disease candidate set in practical clinical studies.

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